



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

1

Homework 5

Department : Computer Science and Engineering

Invigilator's Signature

Total No. of Pages

Mid Semester

End Semester

Make - up

Suppl. Exam

STUDENT'S NAME

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Programme :

BTech

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Course No. :

EE5750

Course Title :

Quantum Error Correction

PART :

Question No.	1	2	3	4	5	6	7	8	9	10	Total
MARKS											

Homework - 5

1.

(a)

$$\Rightarrow |\psi\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle - |11\rangle) \text{ on } C^2 \otimes C^2$$

$$|\psi\rangle = \sum_{i=0}^1 \sum_{j=0}^1 a_{ij} |i\rangle \otimes |j\rangle$$

$$A = (a_{ij})$$

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\Rightarrow A^+ A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \frac{1}{2} I$$

$$\boxed{\mu_1 = \mu_2 = \frac{1}{\sqrt{2}}} \longrightarrow \text{Schmidt coefficients.}$$

$$\boxed{r=2} \longrightarrow \text{Schmidt rank}$$

$$r > 1 \longrightarrow \boxed{\text{Entangled state}}$$

$$\Rightarrow \text{SVD: } C = UDV$$

$$C = \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

orthonormal columns

orthonormal rows.

⇒ Schmidt decomposition:

$$|\psi\rangle = \mu_1 |\psi_1\rangle \otimes |\phi_1\rangle + \mu_2 |\psi_2\rangle \otimes |\phi_2\rangle$$

$$\mu_1 = \frac{1}{\sqrt{2}}; |\psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle; |\phi_1\rangle = |0\rangle$$

$$\mu_2 = \frac{1}{\sqrt{2}}; |\psi_2\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle; |\phi_2\rangle = |1\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} |+\rangle |0\rangle + \frac{1}{\sqrt{2}} |-\rangle |1\rangle$$

(b) ⇒ $|\psi\rangle = \frac{1}{\sqrt{3}} (|00\rangle + |11\rangle + |21\rangle)$ on $\mathbb{C}^3 \otimes \mathbb{C}^2$

$$|\psi\rangle = \sum_{i=0}^2 \sum_{j=0}^1 a_{ij} |i\rangle \otimes |j\rangle$$

$$A = (a_{ij})$$

$$A = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{1}{\sqrt{3}} \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \xrightarrow{a_i} \text{orthogonal}$$

$$\Rightarrow \| \cdot \| = \frac{1}{\sqrt{3}} \quad \Rightarrow \| \cdot \| = \sqrt{\frac{2}{3}}$$

$$\mu_1 = \sqrt{\frac{2}{3}}$$

$$\mu_2 = \frac{1}{\sqrt{3}}$$

Schmidt coefficients.

Schmidt rank

$$r = 2$$

$$r > 1 \rightarrow$$

Entangled state

$$\Rightarrow \text{let } A = U D V^T$$

$$|u_1\rangle = \frac{a_0}{\|a_0\|} = \frac{|1\rangle + |2\rangle}{\sqrt{2}} = |\psi_1\rangle$$

$$|u_2\rangle = \frac{a_0}{\|a_0\|} = |0\rangle = |\psi_2\rangle$$

$$|v_1\rangle = |1\rangle = |\phi_1\rangle \quad |v_2\rangle = |0\rangle = |\phi_2\rangle$$

$$\Rightarrow |\psi\rangle = \sqrt{\frac{2}{3}} \frac{|1\rangle + |2\rangle}{2} \otimes |1\rangle + \frac{1}{\sqrt{3}} |0\rangle \otimes |0\rangle$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 μ_1 $|\psi_1\rangle$ $|\phi_1\rangle$ μ_2 $|\psi_2\rangle$ $|\phi_2\rangle$

Schmidt decomposition.

3. $\Rightarrow$ Alice starts with 2 physical qubits A, B; prepares them in EPR state

$$|EPR\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

She has 2 classical bits b_1, b_2

$$\hookrightarrow |\psi\rangle_{AB} = (X^{b_1} \otimes I)(Z^{b_2} \otimes I) |EPR\rangle$$

$$(a) \Rightarrow (b_1, b_2) = (0, 0) \longrightarrow |\psi\rangle_{AB} = |EPR\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$\Rightarrow (b_1, b_2) = (0, 1) \longrightarrow$$

$$|\psi\rangle_{AB} = (Z \otimes I) |EPR\rangle$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \left(\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \right) = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\Rightarrow (b_1, b_2) = (1, 0)$$

$$|\psi\rangle_{AB} = (X \otimes I) |EPR\rangle = \left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) |EPR\rangle$$

$$= \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$\Rightarrow (b_1, b_2) = (1, 1)$$

$$|\psi\rangle_{AB} = (X \otimes I)(Z \otimes I) |EPR\rangle = \frac{1}{\sqrt{2}} (|10\rangle - |01\rangle)$$

$$\begin{aligned}
 (b) \Rightarrow & \left. \begin{aligned} |\psi_{00}\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ |\psi_{01}\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ |\psi_{10}\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |\psi_{11}\rangle &= \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle) \end{aligned} \right\} |\psi_{b_1 b_2}\rangle \text{ form}
 \end{aligned}$$

$\Rightarrow$ These 4 encoded states are orthonormal in $C^2 \otimes C^2$

$\Rightarrow$ Bell states are pairwise orthonormal

$$\begin{aligned}
 \langle \psi_{b_1 b_2} | \psi_{b'_1 b'_2} \rangle &= \delta_{b_1 b'_1} \delta_{b_2 b'_2} \\
 &= \begin{cases} 1 & \text{if } b_1 = b'_1 \text{ and } b_2 = b'_2 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

(c) $\Rightarrow$ Unitary gate $U \in C^{4 \times 4}$ that maps these 4 states to the computational basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ in some order.

$$\Rightarrow H = \frac{1}{\sqrt{2}}(X + Z)$$

$\hookrightarrow$ Hadamard gate

$$\Rightarrow CNOT(|\psi_{00}\rangle) = CNOT\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right)$$

$$= \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = |+\rangle \otimes |0\rangle$$

$$\Rightarrow CNOT(|\psi_{01}\rangle) = CNOT\left(\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)\right)$$

$$= \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = |-\rangle \otimes |0\rangle$$

$$\Rightarrow CNOT(|\psi_{10}\rangle) = CNOT\left(\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)\right)$$

$$= \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) = |+\rangle \otimes |1\rangle$$

$$\Rightarrow \text{CNOT}(|\psi_{11}\rangle) = \text{CNOT}\left(\frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)\right) \\ = \frac{1}{\sqrt{2}}(|11\rangle - |00\rangle) = |1-\rangle \otimes |1\rangle$$

$$H(|+\rangle) = |0\rangle, \quad H(|-\rangle) = |1\rangle$$

$$\Rightarrow U = (H \otimes I) \cdot \text{CNOT}_{A \rightarrow B}$$

$$\Rightarrow \text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

$$\Rightarrow U|\psi_{00}\rangle = |00\rangle, \quad U|\psi_{01}\rangle = |01\rangle$$

$$U|\psi_{10}\rangle = |10\rangle, \quad U|\psi_{11}\rangle = |11\rangle$$

$$(d) \Rightarrow \forall |\psi_{b_1 b_2}\rangle, \quad (b_1 b_2 = 00, 01, 10, 11)$$

$$\rho_A = \text{Tr}_B(|\psi_{b_1 b_2}\rangle\langle\psi_{b_1 b_2}|)$$

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|1\rangle)$$

$$|\psi_{01}\rangle = \frac{1}{\sqrt{2}}(|0\rangle|0\rangle - |1\rangle|1\rangle)$$

$$|\psi_{10}\rangle = \frac{1}{\sqrt{2}}(|0\rangle|1\rangle + |1\rangle|0\rangle)$$

$$|\psi_{11}\rangle = \frac{1}{\sqrt{2}}(|0\rangle|1\rangle - |1\rangle|0\rangle)$$

$$\rho_A = \frac{1}{2}|\psi_0\rangle\langle\psi_0| + \frac{1}{2}|\psi_1\rangle\langle\psi_1|$$

$$\rho_A = \frac{1}{2}|\psi_0\rangle\langle\psi_0| + \frac{1}{2}|\psi_1\rangle\langle\psi_1| = \frac{1}{2}I$$

Schmidt decomposition

$$|\psi\rangle_{AB} = \sum_{i=0}^{\infty} \sqrt{p_i} |\phi_i\rangle_A \otimes |\chi_i\rangle_B$$

$$\rho_A = \sum_i p_i |\phi_i\rangle\langle\phi_i|_A$$

$$11 \Rightarrow S_B = \frac{1}{2} |\phi_1\rangle\langle\phi_1| + \frac{1}{2} |\phi_2\rangle\langle\phi_2|$$

$$S_A = S_B = \frac{1}{2} I$$

So in all 8 cases — Eve intercepts A or B,
for any (b_1, b_2) — Eve's qubit is always
in same state $S = \frac{1}{2} I$

(c) $\Rightarrow$ Eve's intercepted qubit is always in state $\frac{1}{2} I$,
independent of b_1, b_2 . So, every possible
measurement outcome has probabilities independent
of encoded bits. Eve can gain no info
about (b_1, b_2) .
 ~~$P(\text{outcome} | b_1, b_2)$~~

This is case of indistinguishable preparations.
 $\rightarrow$ Even if global states are orthogonal,
Eve's local reduced state is identical for all
choices of (b_1, b_2) .

(f) $\Rightarrow$ Alice uses deterministic function
 $(b_1, b_2) \mapsto (A, B)$

Bob must be able to recover (b_1, b_2) from
 (A, B) . Map must be injective. As there are
4 possible input pairs and 4 possible output
pairs, 4 output pairs (A, B) are distinct.

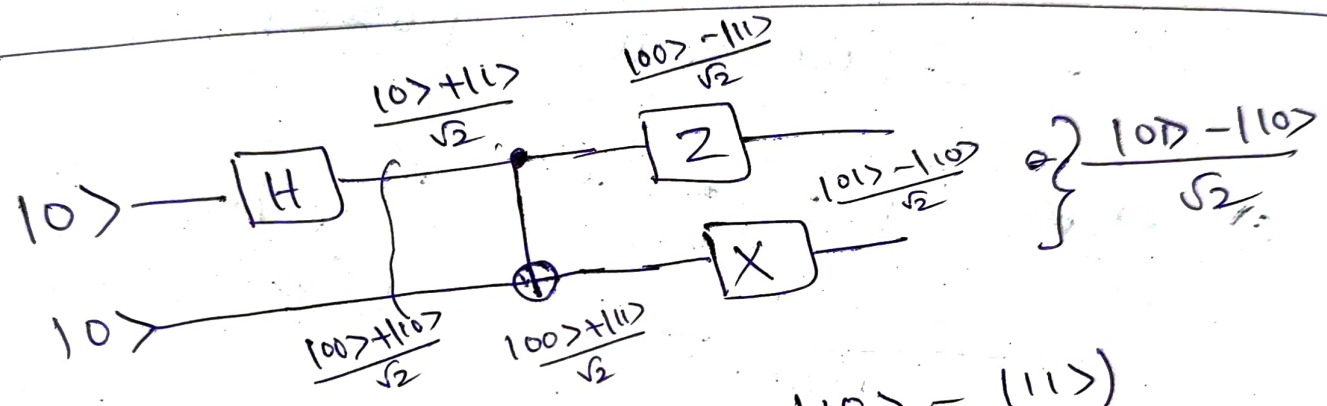
(g) $\Rightarrow$ 4 output pairs (A, B) are distinct.
1st bit A can't be constant. (then only 2 output pairs)
~~A is not~~ A is not constant, B is also
not constant

⇒ If Eve intercepts either classical bit, she obtains bit that is correlated with (b_1, b_2) . So, gains some info about (b_1, b_2) .

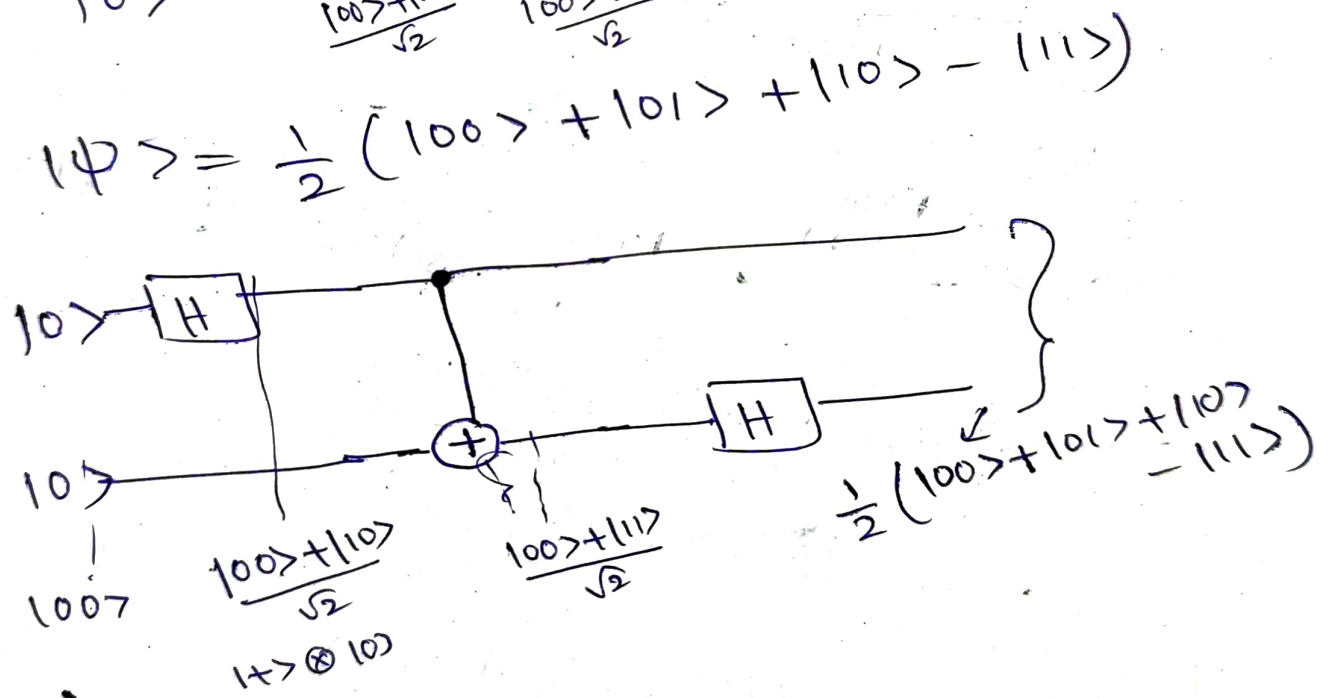
$$I((b_1, b_2); A) = H(A) > 0$$

|| by for B. → Classical scheme always makes Eve gather some info about (b_1, b_2)

4.
(a)



(b)



Explanation:

$$\begin{aligned}
 &100\rangle \xrightarrow{H} H|00\rangle = \frac{100\rangle + 110\rangle}{\sqrt{2}} \\
 &\xrightarrow{\text{CNOT}_{1,2}} \frac{1}{\sqrt{2}}(100\rangle + 111\rangle) \xrightarrow{Z \text{ on qubit 1}} \frac{1}{\sqrt{2}}(100\rangle - 111\rangle)
 \end{aligned}$$

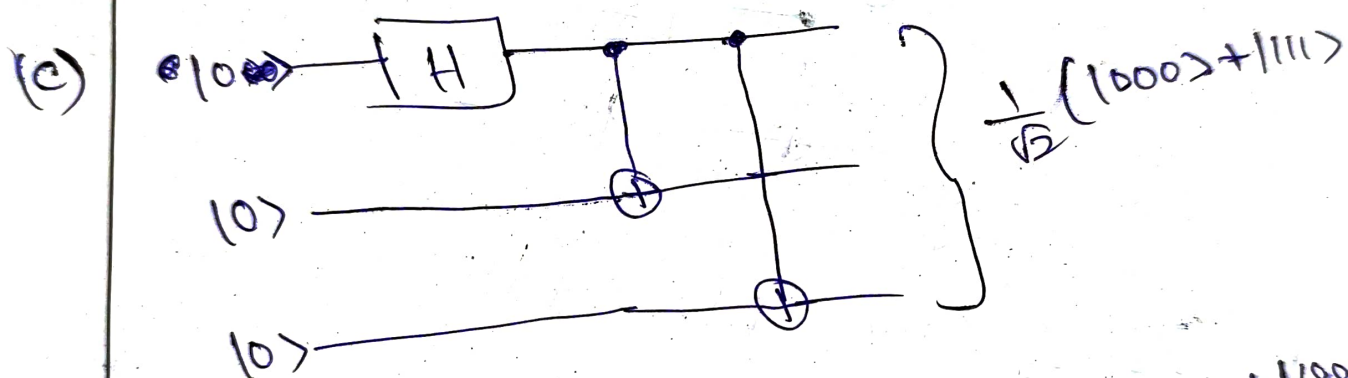
$$\xrightarrow{X \text{ on qubit 2}} \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

(b) Explanation: H on qubit 1 $\rightarrow \frac{(|0\rangle + |1\rangle)}{\sqrt{2}} \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$

$$\xrightarrow{CNOT_{1,2}} \frac{|00\rangle + |11\rangle}{\sqrt{2}} \xrightarrow{H \text{ on qubit 2}}$$

$$\frac{1}{\sqrt{2}} \left(|0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} + |1\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle - |11\rangle)$$



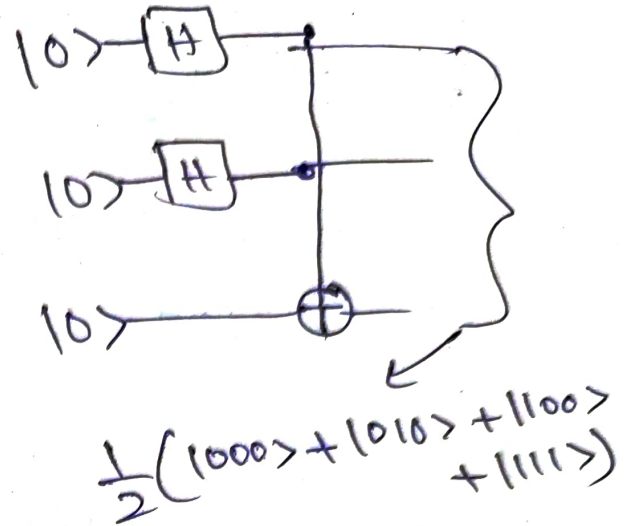
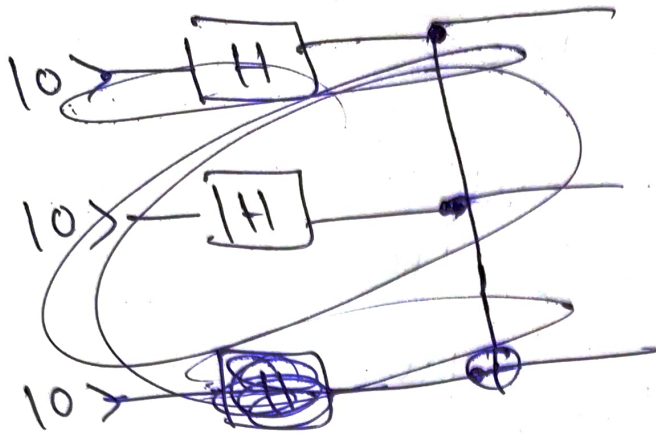
$$|000\rangle \xrightarrow{H \text{ on qubit 1}} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |00\rangle = \frac{|000\rangle + |100\rangle}{\sqrt{2}}$$

$$\xrightarrow{CNOT_{1,2}} \frac{1}{\sqrt{2}} \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) \otimes |0\rangle$$

$$= \frac{1}{\sqrt{2}} (|000\rangle + |110\rangle)$$

$$\xrightarrow{CNOT_{1,3}} \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)$$

d)



$$|1000\rangle \xrightarrow{\text{H on qubit 1}} \left(\frac{|10\rangle + |11\rangle}{\sqrt{2}} \right) \otimes |10\rangle \otimes |10\rangle$$

$$\xrightarrow{\text{H on qubit 2}} \left(\frac{|10\rangle + |11\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|10\rangle + |11\rangle}{\sqrt{2}} \right) \otimes |10\rangle$$

$$\text{Toffoli} \Rightarrow \frac{1}{2}(|1000\rangle + |010\rangle + |100\rangle + |110\rangle)$$

$$\text{Toffoli} \Rightarrow \frac{1}{2}(|1000\rangle + |010\rangle + |1010\rangle + |111\rangle)$$

Grates operation:

$$\Rightarrow I|10\rangle = |10\rangle ; I|11\rangle = |11\rangle$$

$$\Rightarrow X|10\rangle = |11\rangle ; X|11\rangle = |10\rangle$$

$$\Rightarrow Y|10\rangle = i|11\rangle ; Y|11\rangle = -i|10\rangle$$

$$\Rightarrow Z|10\rangle = |10\rangle ; Z|11\rangle = -|11\rangle$$

$$\Rightarrow H|10\rangle = \frac{|10\rangle + |11\rangle}{\sqrt{2}} = |1+\rangle ; H|11\rangle = \frac{|10\rangle - |11\rangle}{\sqrt{2}} = |1-\rangle$$

$$\Rightarrow \text{CNOT}_{1,2} (100\rangle = 100\rangle$$

$$\text{CNOT}_{1,2} (101\rangle = 101\rangle$$

$$\text{CNOT}_{1,2} (110\rangle = 111\rangle$$

$$\text{CNOT}_{1,2} (111\rangle = 110\rangle$$

$$\Rightarrow \text{Toffoli} (1000\rangle = 1000\rangle$$

$$\text{Toffoli} (1001\rangle = 1001\rangle$$

$$\text{Toffoli} (1010\rangle = 1010\rangle$$

$$\text{Toffoli} (1011\rangle = 1011\rangle$$

$$\text{Toffoli} (1100\rangle = 1100\rangle$$

$$\text{Toffoli} (1101\rangle = 1101\rangle$$

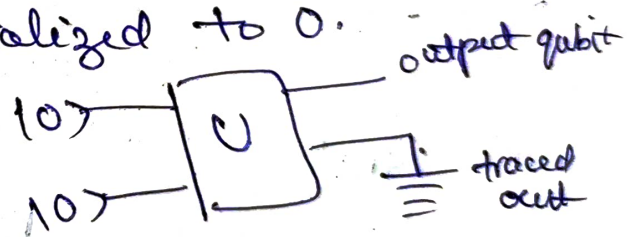
$$\text{Toffoli} (1110\rangle = 1111\rangle$$

$$\text{Toffoli} (1111\rangle = 1110\rangle$$

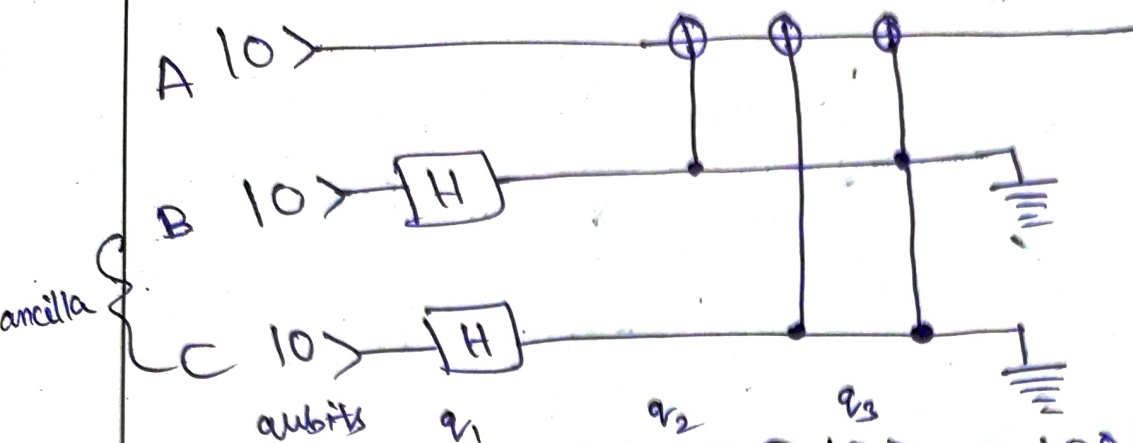
5.

$\Rightarrow$ Use composite system of 1 or more qubits.

All qubits must be initialized to 0.



(a) $\Rightarrow$ A classical RV that can take 2 possible values 0 and 1, with prob. $\frac{1}{4}$, $\frac{3}{4}$ respectively.



$$|\psi\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle = |000\rangle$$

↓ H-gate to q_2

$$|\psi\rangle = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |010\rangle)$$

↓ H-gate to q_3

$$|\psi\rangle = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2} (|000\rangle + |001\rangle + |010\rangle + |011\rangle)$$

~~Facing CNOT~~
↓ CNOT_{B→A}

$$|\psi\rangle = \frac{1}{2} (|000\rangle + |001\rangle + |110\rangle + |111\rangle)$$

↓ CNOT_{C→A}

$$|\psi\rangle = \frac{1}{2} (|000\rangle + |101\rangle + |110\rangle + |011\rangle)$$

↓ Toffoli

$$|\psi\rangle = \frac{1}{2} (|000\rangle + |101\rangle + |110\rangle + |011\rangle)$$

↓ Trace out wrt B

$$\rho_{AC} = \frac{1}{4} (|00\rangle + |11\rangle) (\langle 00| + \langle 11|) + \frac{1}{4} (|10\rangle + |11\rangle) (\langle 10| + \langle 11|)$$

$$[\rho_{AC} = \langle 0|_B \rho_{ABC} |0\rangle_B + \langle 1|_B \rho_{ABC} |1\rangle_B]$$

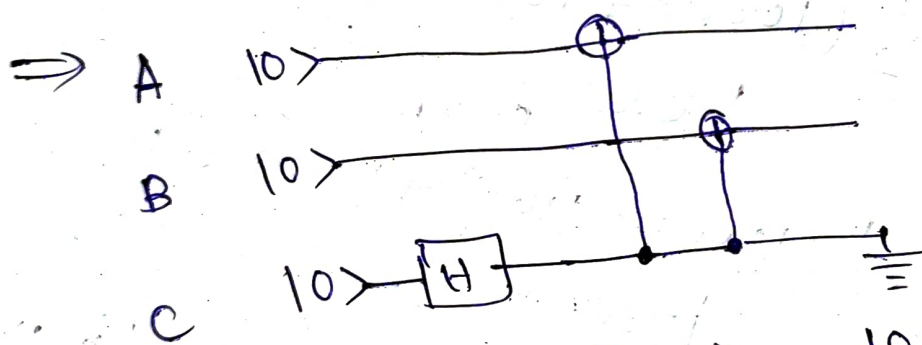
Tracing out wire C

$$\rho_A = \langle 0|_C \rho_{AC} |0\rangle_C + \langle 1|_C \rho_{AC} |1\rangle_C$$

$$\rho_A = \frac{1}{4} |0\rangle\langle 0| + \frac{3}{4} |1\rangle\langle 1|$$

$$\rho_A = \begin{bmatrix} 1/4 & 0 \\ 0 & 3/4 \end{bmatrix}$$

(b) $\Rightarrow$ 2 classical bits A and B that are each uniformly random but always equal. Pair (A, B) takes value (0, 0) with prob. $\frac{1}{2}$ and (1, 1) with prob. $\frac{1}{2}$, does not take values (0, 1), (1, 0)



$$|\psi\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle = |000\rangle$$

$\downarrow$ H-gate on C

$$|\psi\rangle = |0\rangle \otimes |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} (|000\rangle + |001\rangle)$$

$\downarrow$ CNOT_{C $\rightarrow$ A}

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|100\rangle + |101\rangle)$$

↓ CNOT_{C→B}

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|1000\rangle + |1111\rangle)$$

↓ Tracing out wrt C

$$\rho_{ABC} = \frac{1}{2} (|1000\rangle + |1111\rangle)(\langle 0000| + \langle 1111|)$$

$$\rho_{AB} = \langle 0|_C \rho_{ABC} |0\rangle_C + \langle 1|_C \rho_{ABC} |1\rangle_C$$

$$\rho_{AB} = \frac{1}{2} (|100\rangle)(\langle 000|) + \frac{1}{2} (|111\rangle)(\langle 111|)$$

$$\rho_{AB} = \frac{1}{2} |100\rangle\langle 000| + \frac{1}{2} |111\rangle\langle 111|$$

THE END